BBP

RASD & DD Presentation

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Politecnico di Milano

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POLITECNICO
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Outline

1. RASD
   - Introduction
   - Overall Description
   - Specific Requirements
   - Alloy
2. DD
   - Architectural Design
   - Implementation, Integration and Test Plan

1 RASD
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2 DD

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Goals

Goal	Description
G1	Recognize suitable bike paths
G2	Allow the user to browse different bike paths
G3	Allow the user to see bike paths statistics
G4	Allow the user to search for paths between two locations
G5	Allow the user to see paths optimality based on other registered users' contributions
G6	Allow user to see weather information when available
G7	Allow users to register to the application using credentials
G8	Merge different registered users' contributions of the same bike path
G9	Allow registered user to confirm or reject false positives in track recording
G10	Allow registered user to let the system automatically record suitable bike paths
G11	Allow registered user to manually create suitable bike paths
G12	Allow registered user to publish bike paths
G13	Allow the store of bike paths data for registered users
G14	Allow registered user to add publishable information about bike paths

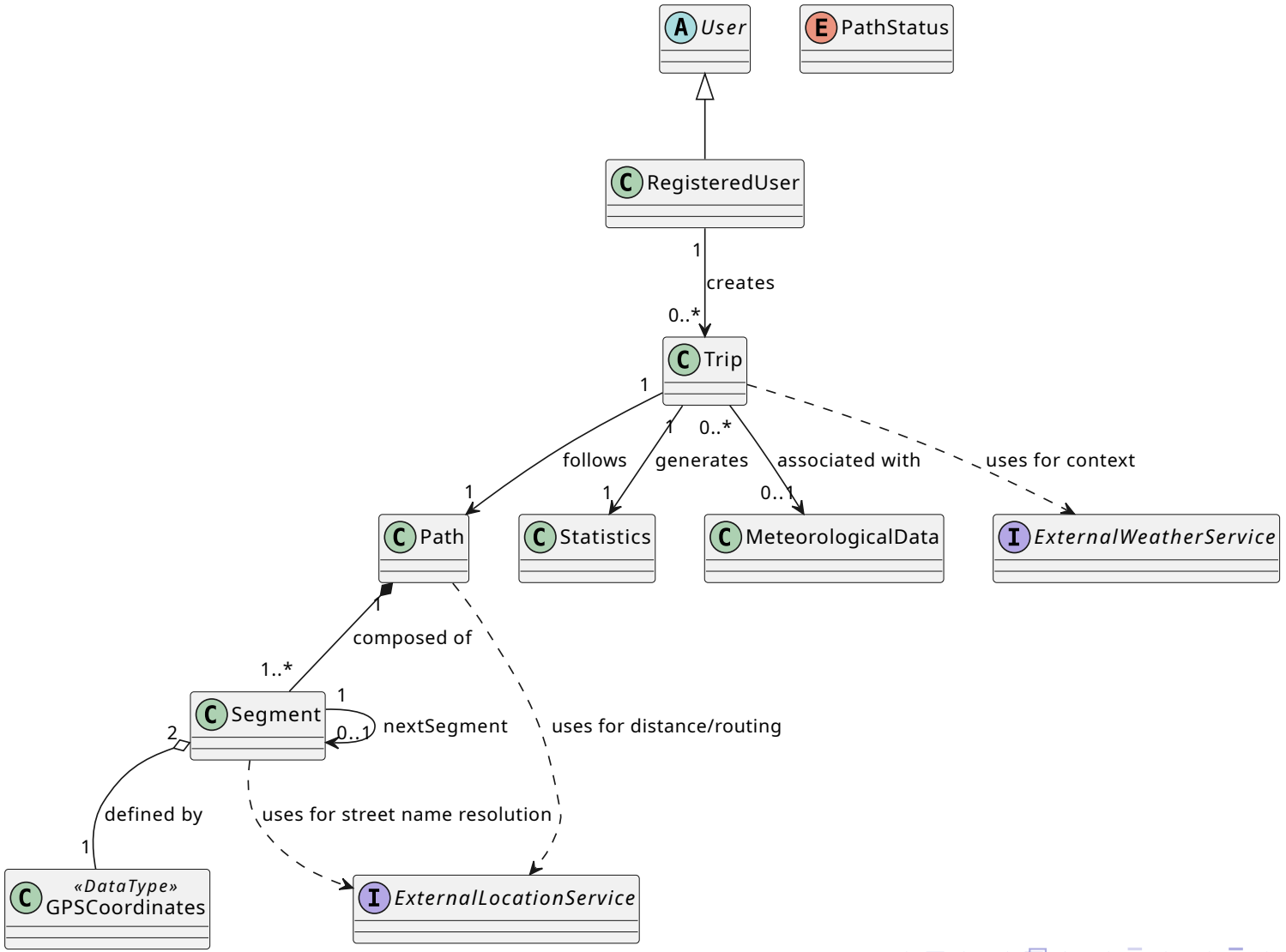
Phenomena

- World Phenomena (WP):
  - *Physical movement*
  - *Road status changes (potholes)*
  - *Weather conditions*
- Shared Phenomena (SP):
  - World-controlled:
    - *Sensors transmitting*
    - *User inputs*
  - Machine-controlled:
    - *Map visualization*
    - *Confirmation requests*

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Class diagram

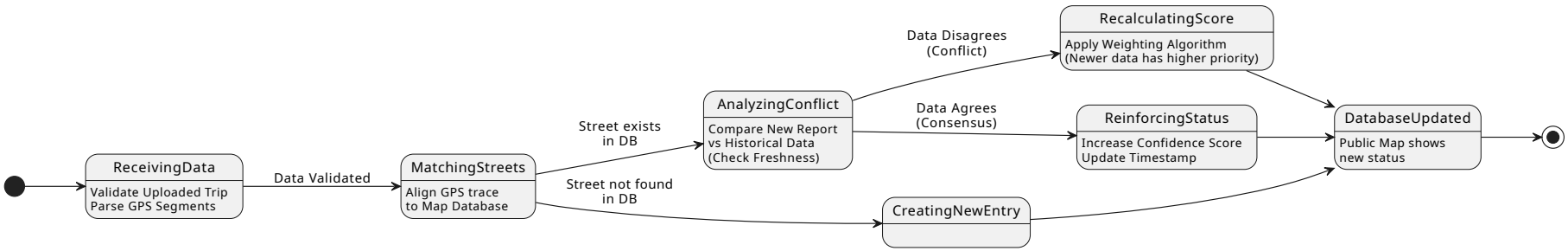


System Dynamics & Data Lifecycle

- **Acquisition Strategy**: hybrid model using **inference** (speed thresholds) and **topological reconstruction** (manual)
- **Validation Pattern**: sensor anomalies are flagged as "**candidates**" to filter **false positives** via user confirmation
- **Consistency Logic**: server-side **weighting algorithm** resolves conflicts based on data freshness and consensus

State Diagram Example

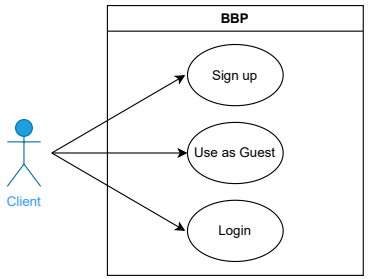
Path Aggregation

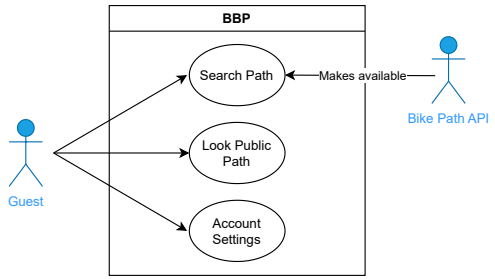


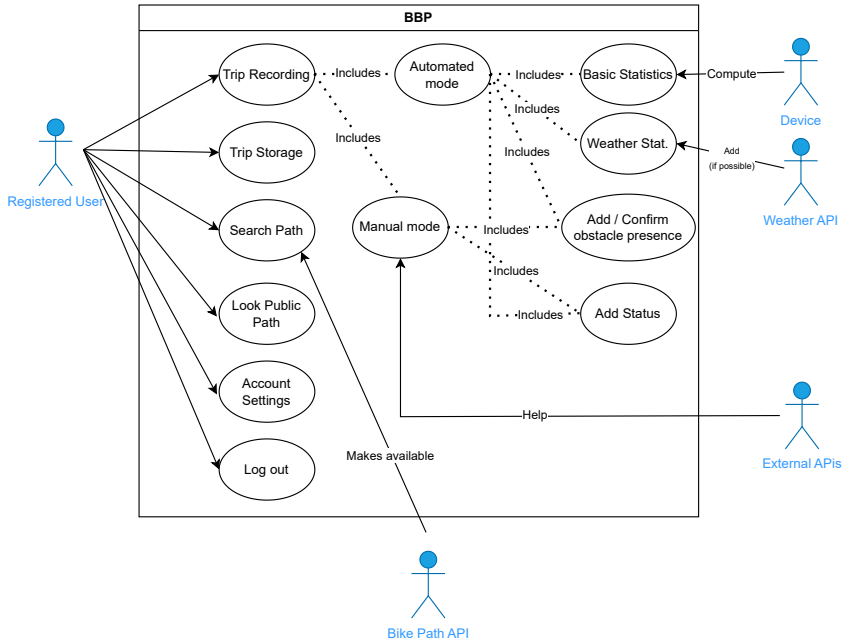
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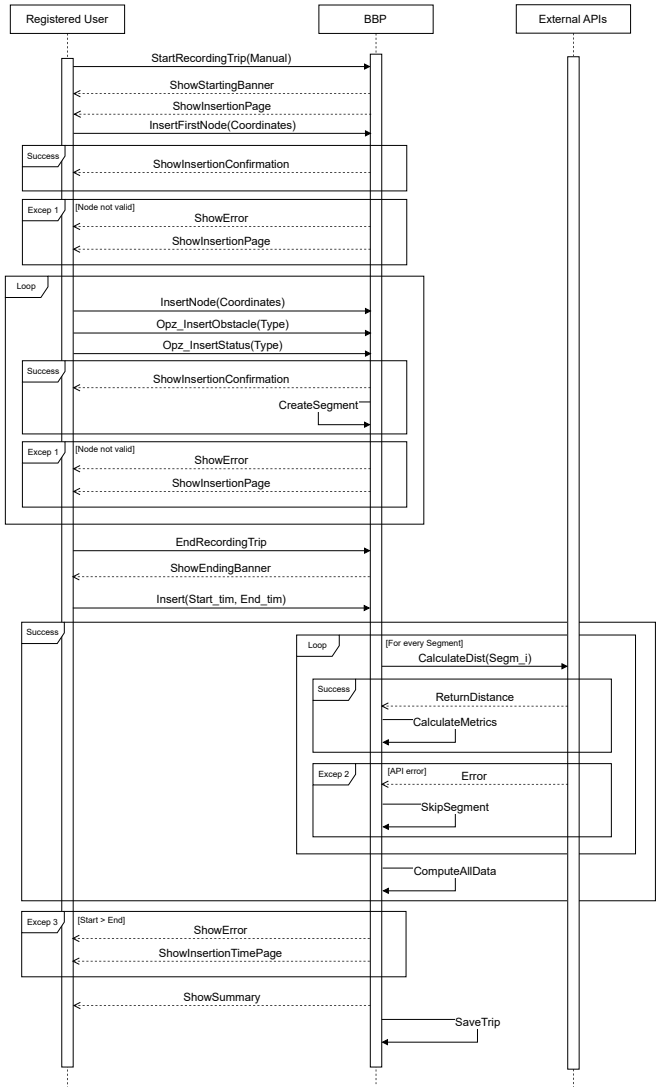
Specific Requirements







Specific Requirements



Specific Requirements

- **Realibility**
- **Availability**
- **Security**
- **Maintainability**
- **Portability**

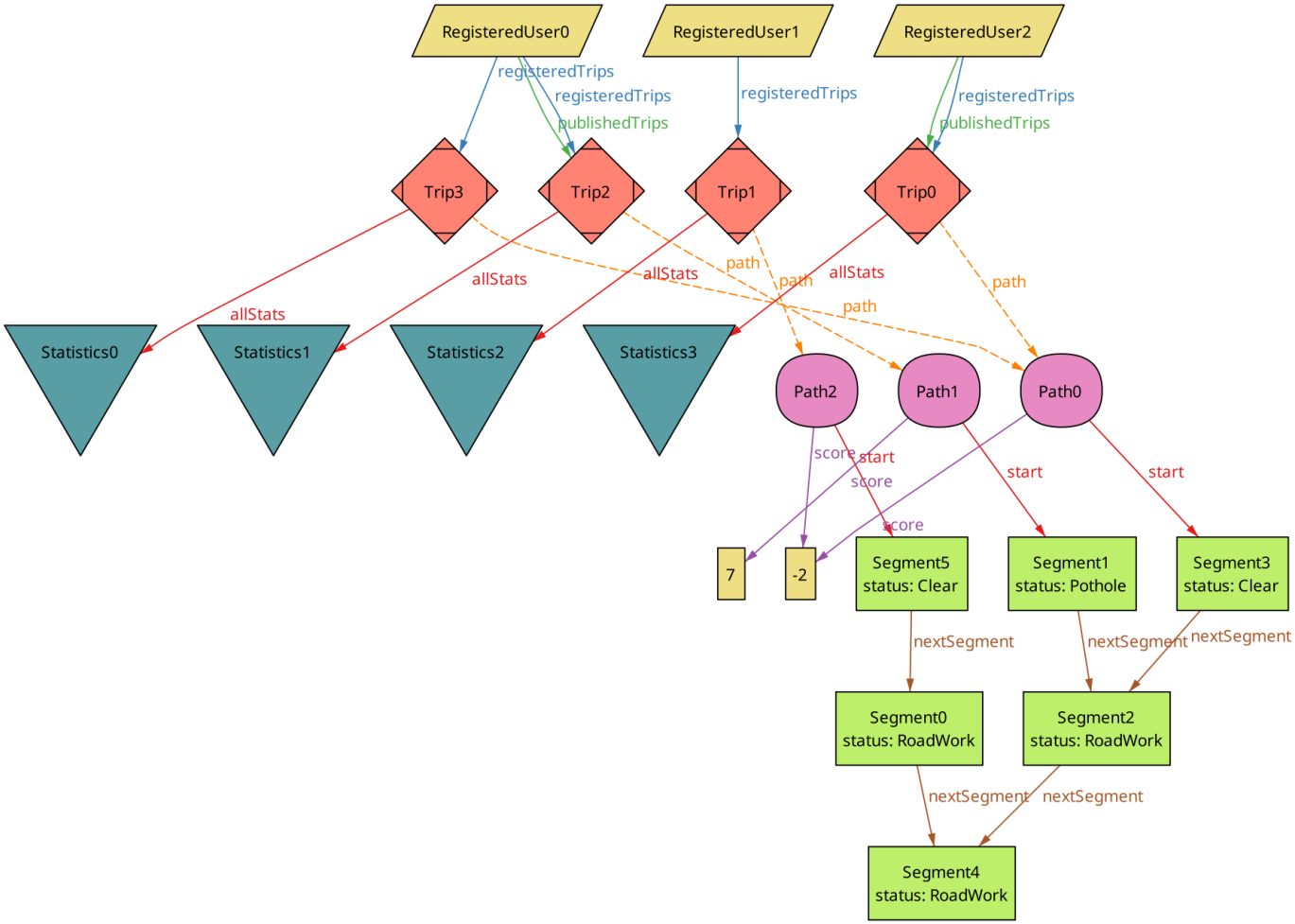
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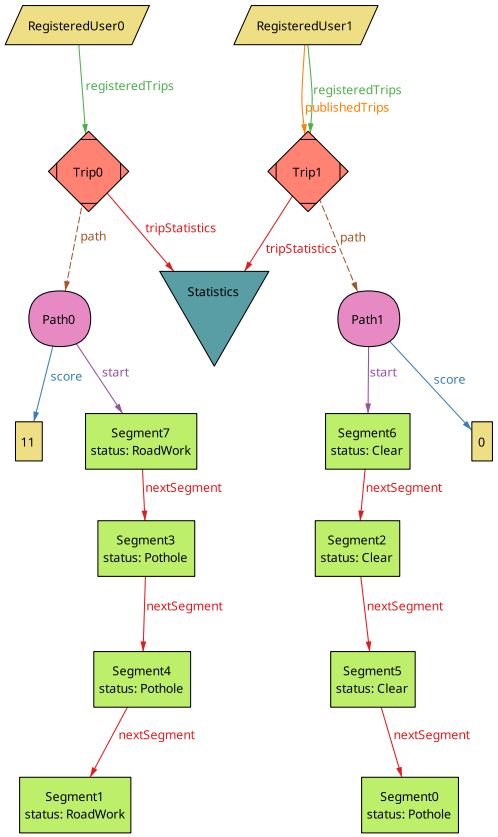
Alloy

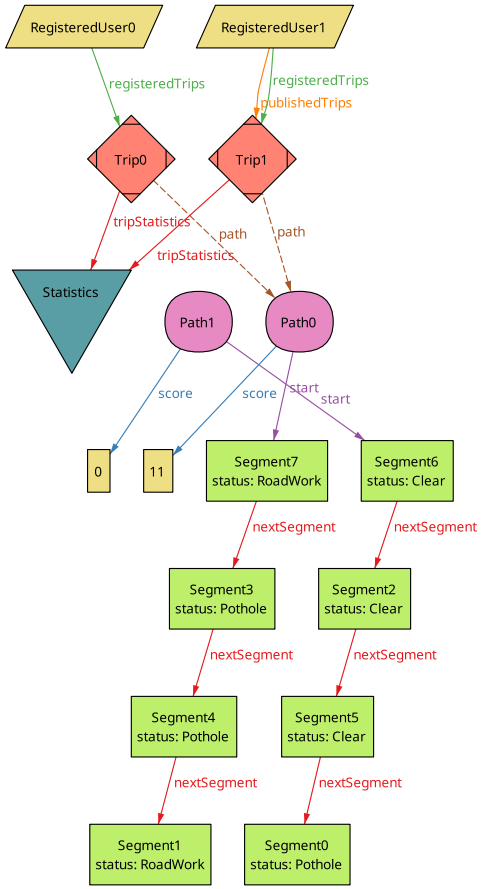
- **Static model analysis** - Understand the logic inside the model itself
- **Merging of two paths** - Proving merging consistency

Static Model Analysis



Merging of Two Paths





1 RASD

2 DD

Architectural Design

Implementation, Integration and Test Plan

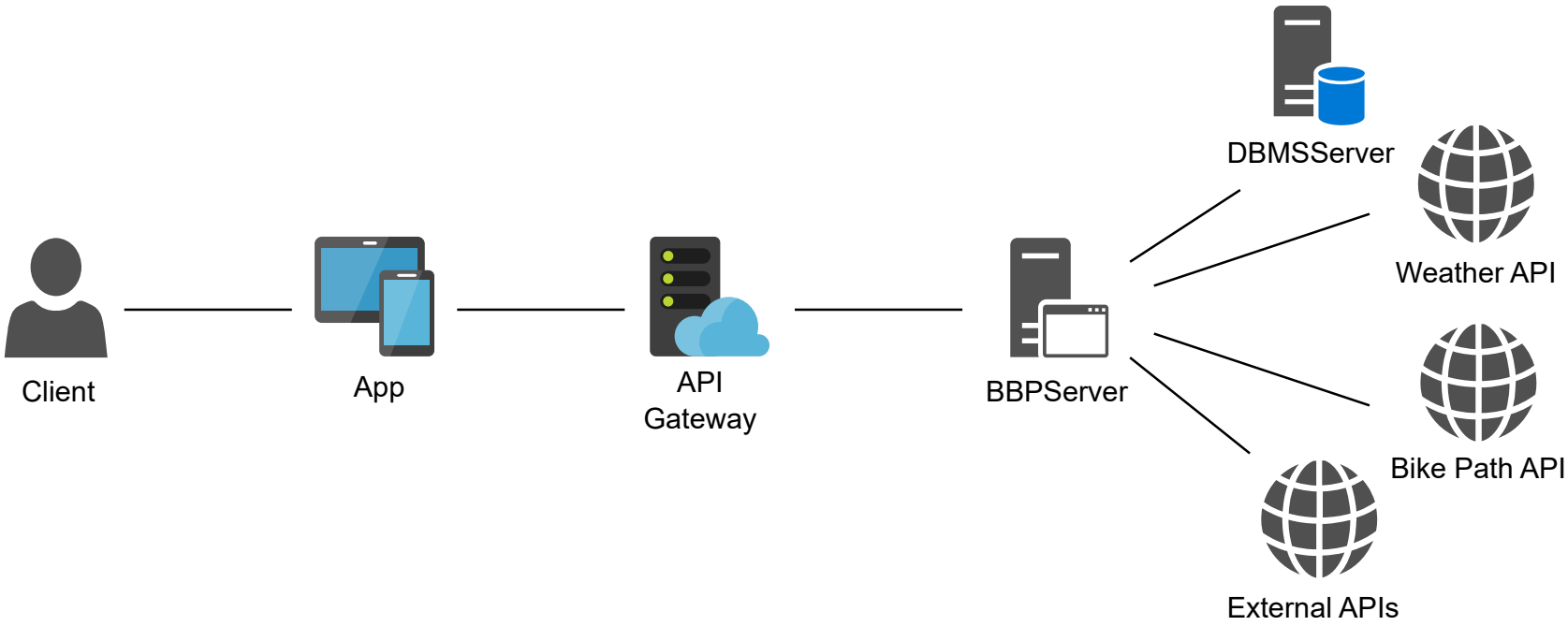
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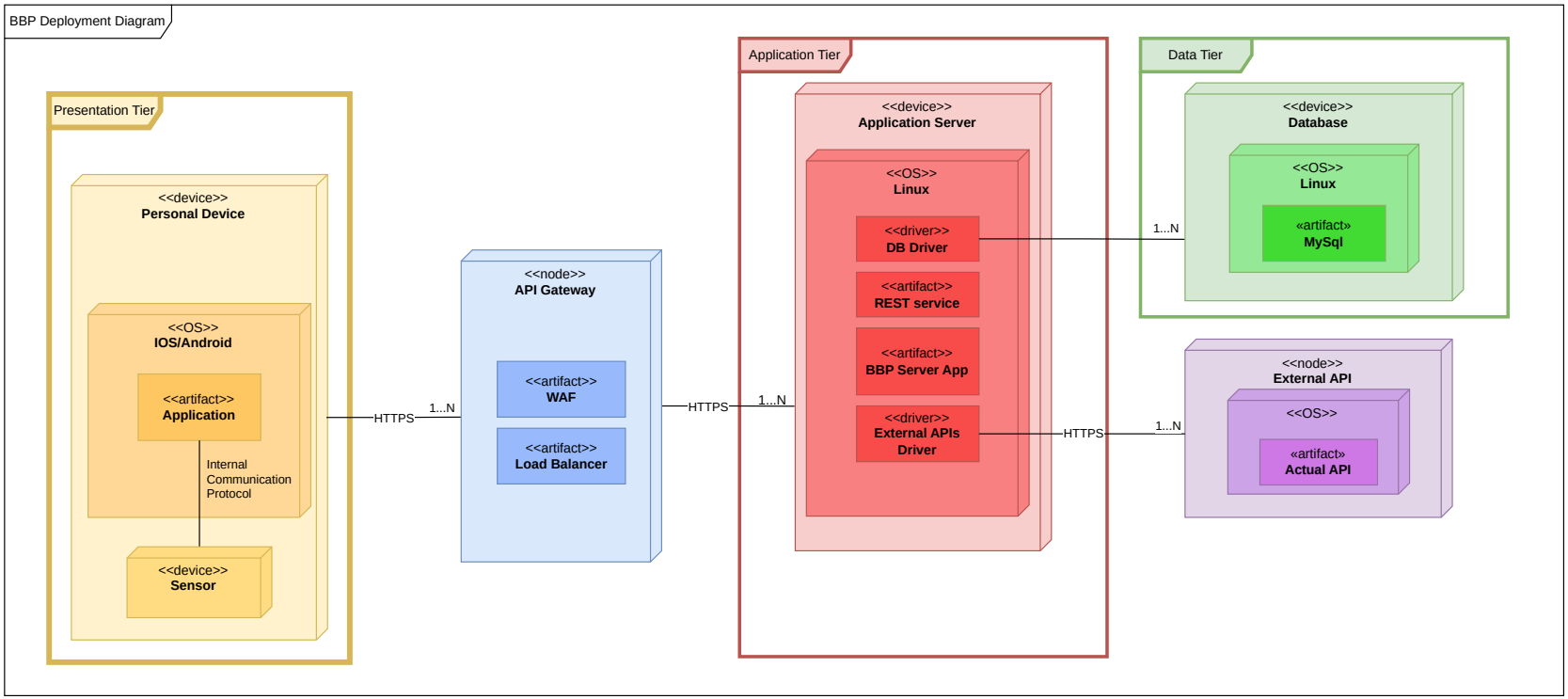
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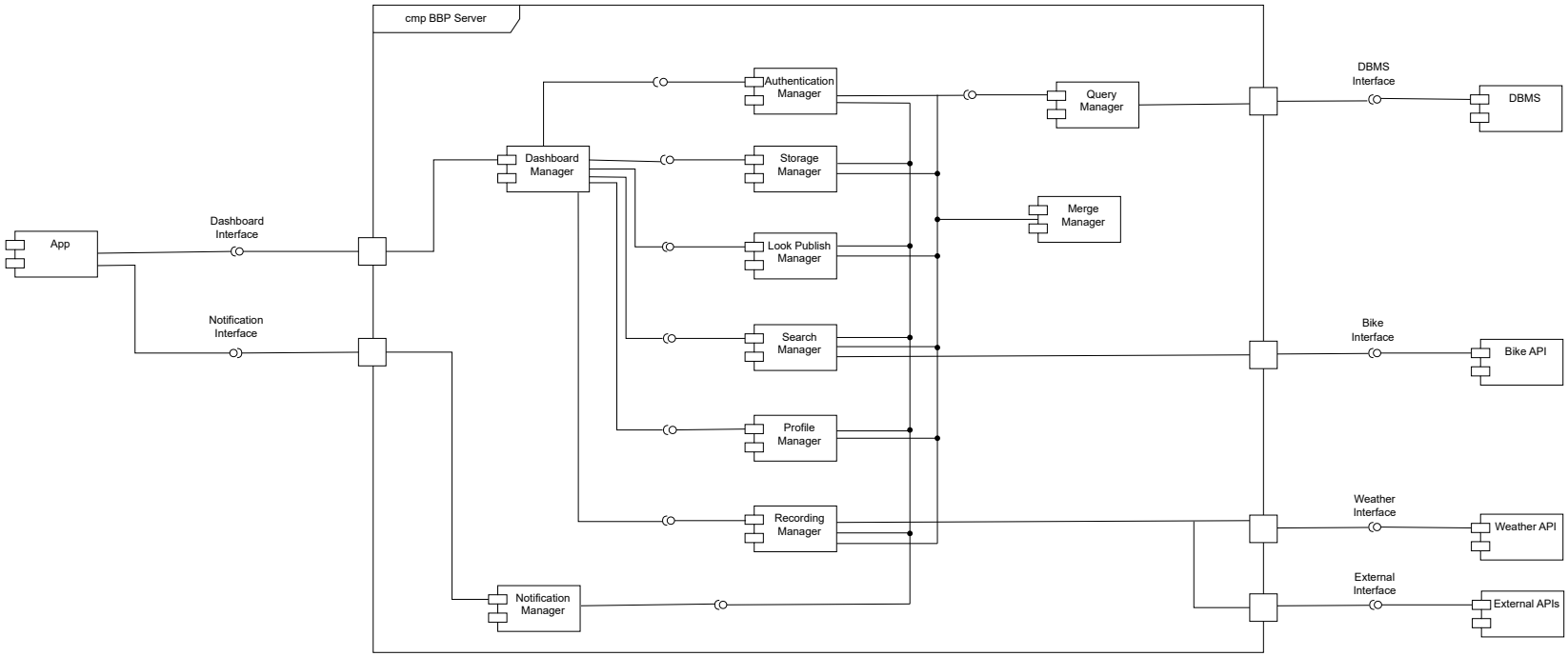
Architectural Design



Deployment View



Architectural Design



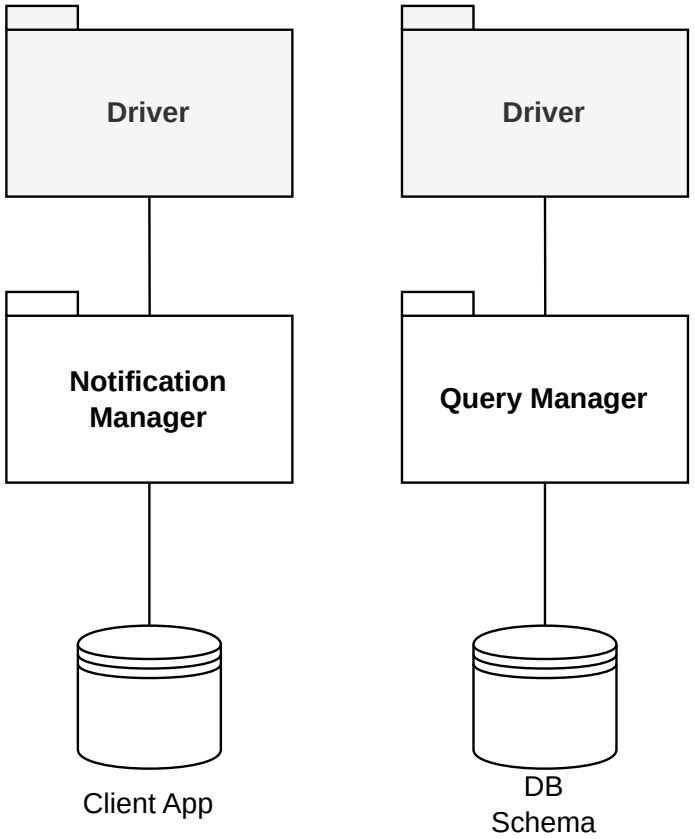
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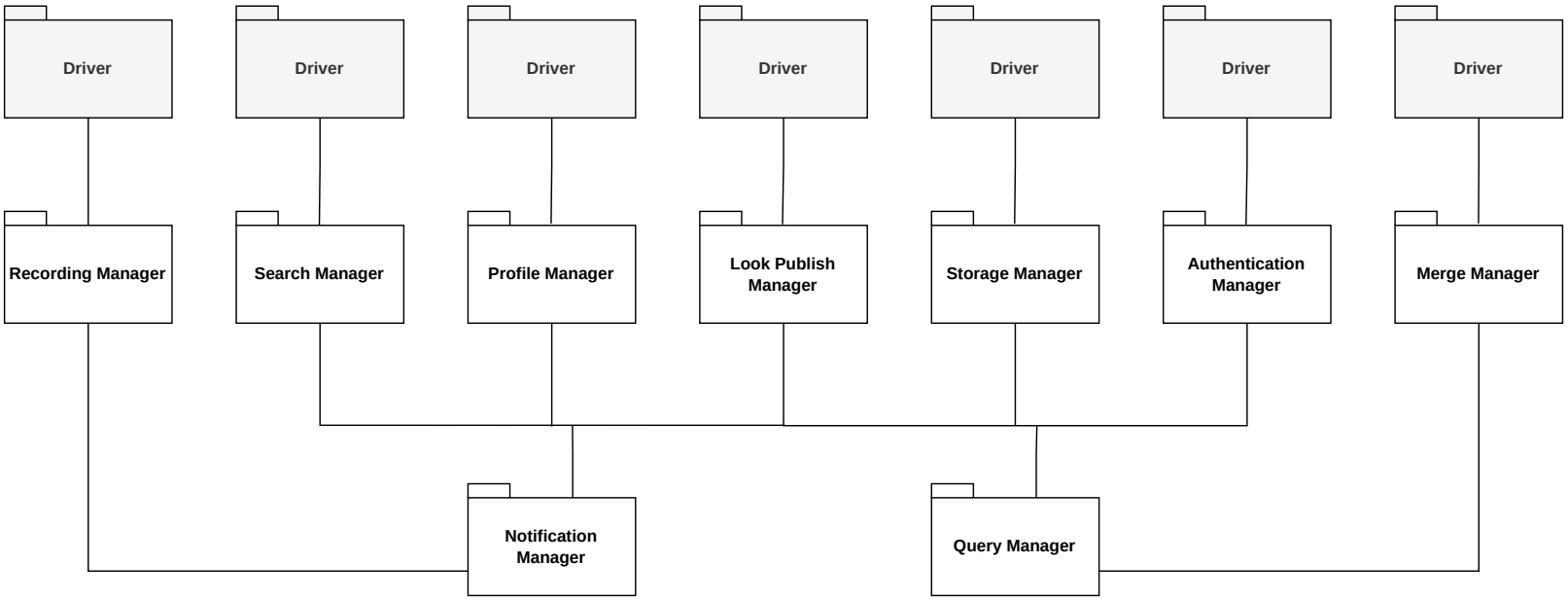
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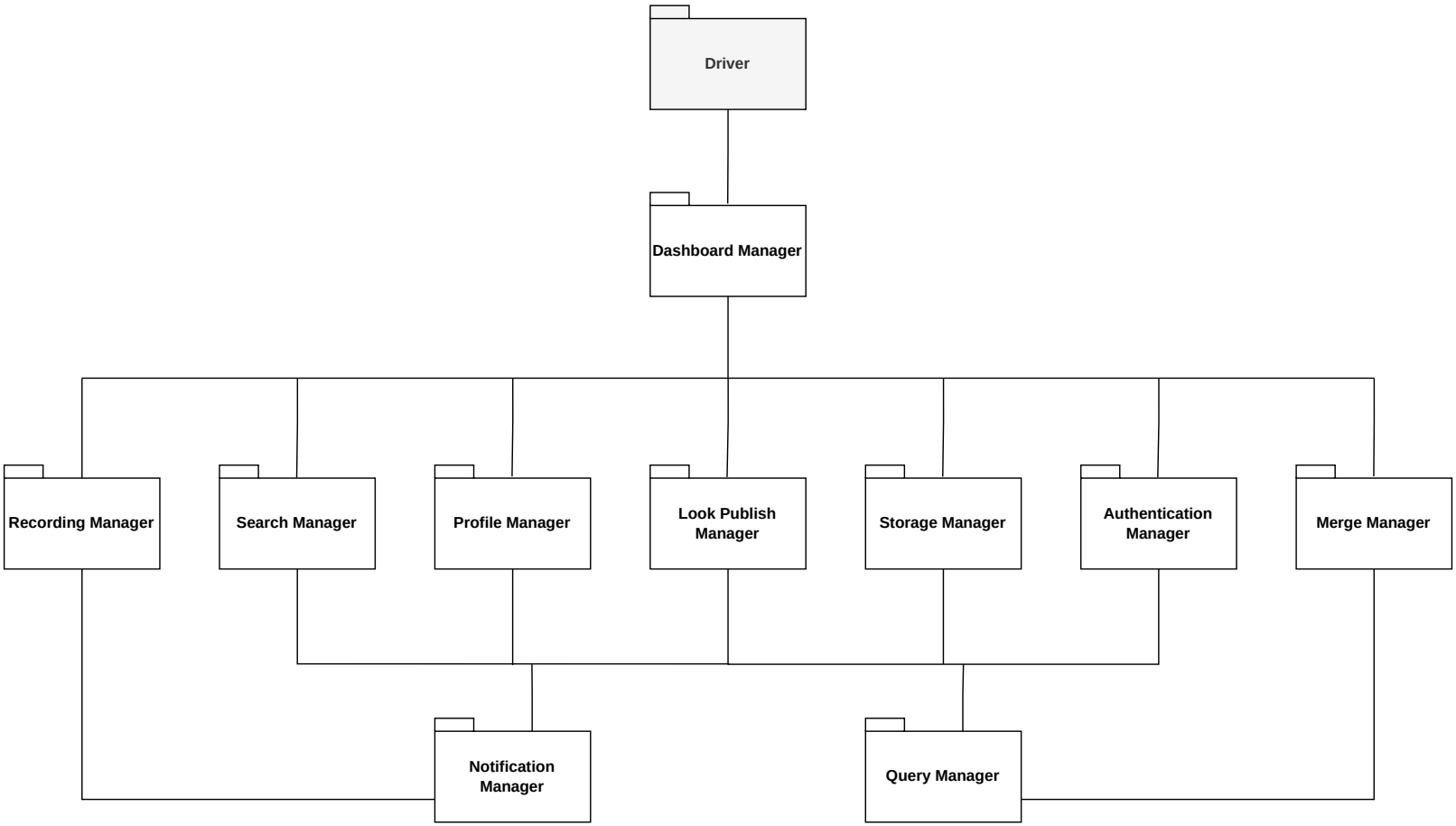
Query and Notification Manager



Parallel Integration



Dashboard Manager



System Testing Strategy

- **Functional Testing**
- **Load Testing**
- **Performance Testing**
- **Stress Testing**
- **User Interface Testing**

Thank you!